papergold
Date: / /

Day 24 , 13/11/26

LeetCode 20 - Valid Parentheses

Approach:-
Use a stack because brackets must close in the reverse order in which they are opened.

Steps

1. If the character is an opening bracket (, {, [, push it into the stack.
2. If it is a closing bracket
   - Check whether the stack is empty - invalid.
   - check whether the top opening bracket matching it.
   - If not - invalid.
   - Otherwise, pop it.
3. At the end, the stack must be empty.

Time Complexity : O(n)
Space Complexity : O(n)

Code :-

```
class Solution {
    public boolean isValid(String s) {
        Stack<Character> stack = new Stack<>();
        for(char ch : s.toCharArray()) {
            if (ch == '(' || ch == '{' || ch == '[') {
                stack.push(ch);
            } else {
                if (stack.isEmpty()) return false;
                char top = stack.pop();
                if((ch == ')' && top != '(') || (ch == '}'
                && top != '{') || (ch == ']' && top != '['))
                return false;
            }
        }
        return stack.isEmpty();
    }
}
```

Problem List

Description | Accepted | Editorial | Solutions | Submissions

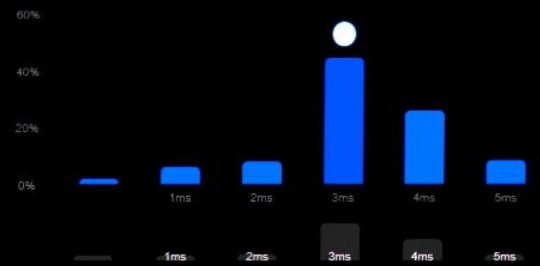
All Submissions

Runtime

3 ms | Beats **85.10%**

Memory

42.85 MB | Beats 83.57%



Code | Java

Submit

Premium

Code

Java | Auto

```
class Solution {
    public boolean isValid(String s) {
        Stack<Character> stack = new Stack<>();
        for (char ch : s.toCharArray()) {
            if (ch == '(' || ch == '{' || ch == '[') {
                stack.push(ch);
            }
            else {
                if (stack.isEmpty()) {
                    return false;
                }
                char top = stack.pop();

                if ((ch == ')' && top != '(') ||
                    (ch == '}' && top != '{') ||
                    (ch == ']' && top != '[')) {
                    return false;
                }
            }
        }
        return stack.isEmpty();
    }
}
```

Saved

Ln 3, Col 48

Testcase | Test Result

Accepted Runtime: 0 ms